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RESEARCH ARTICLE

Lifelogging By Senior Citizens: Implications from a Light-Weight GPS-Based Study

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Abstract

Background: The World Health Organization (WHO) reported that the proportion of senior citizens (over 60 years old) in the world population will reach 30% by 2050. Governments with rapidly ageing societies, such as Japan, urge local communities to develop sustainable solutions that facilitate healthy ageing. A vital component of the solution is for individuals to self-monitor their daily activities and health statuses. With advances in wearable devices, the self-monitoring practice of lifelogging, has become accessible for many people, making such devices promising tools for senior citizens. However, the current understanding of the effective practices of supporting senior citizens through lifelogging technologies is still limited.

Methods: This article reports the findings of a study that investigated the feasibility and effectiveness of a lightweight GPS-based lifelogging approach exercised by senior citizens in Japan. We asked ten participants to carry a smartphone when they went out, and the device automatically captured their activity and location data. We also generated monthly personalised reports to help participants reflect on their daily lives. We analysed the log data collected by the mobile devices, answers to questionnaires distributed every month with activity reports, and final interviews.

Results: The results of the analysis suggest that 1) it is feasible for senior citizens to carry a smartphone to collect their activity and location data, and participants did not feel stressed even when they did not have previous experience using the device; 2) activity reports are a promising way to help senior citizens reflect on their daily lives; 3) senior citizens can be highly atuned to erroneous numeric data captured by these devices, and 4) the ability to interpret visualisations of lifelog data can vary across participants.

Conclusions: The findings of this study call for developing data literacy programs for senior citizens to facilitate their effective use of data-driven services.

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Any reports and responses or comments on the article can be found at the end of the article.

Keywords
Healthy Aging, Self-Monitoring, Senior Citizens, Lifelogging, Wearable Devices, Smart Devices

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Introduction

Population ageing is a global phenomenon. The pace of population ageing has occurred fastest in Japan according to both the conventional and the economic old-age dependency ratio (United Nations, 2015; 2020; Statistics Bureau, 2020). Local communities are expected to develop sustainable solutions that facilitate healthy ageing, which is defined as "the process of developing and maintaining the functional ability that enables well-being in older age" (World Health Organization, 2015). Individuals can improve their health and well-being through lifelong health promotion and preventive care to maintain maximum functional capacity. In addition, the health and long-term care system must be tailored to meet the needs of the ageing population by providing age-appropriate integrated care and focusing on maintaining internal capacity (United Nations, 2020). Assistive lifelogging technologies have the potential to support older, frailer people, as well as caregivers, in their everyday lives (Offermann-van Heek et al., 2020). In addition, wearable devices that support lifelogging have been reported to involve global spending of $81.5bn in 2021 and this sum is expected to grow further according to a recent market report (Gartner, 2021). The commodification of wearable devices and smart devices provides a great opportunity to extend lifelogging experience to a wide range of populations including senior citizens.

Lifelogging is defined as "a form of pervasive computing, consisting of a unified digital record of the *totality* of an individual's experiences, captured multi-modally through digital sensors and stored permanently as a personal multi-media archive" (Dodge and Kitchin, 2007, p. 431). Sellen and Whittaker (2010) proposed use cases of such personal multimedia archives in terms of the five Rs: recollecting, reminiscing, retrieving, reflecting and remembering intentions. Of those, reflection has been recognised as one of the most important skills and processes by professionals (Schön, 1984). Although systematic reflection has been studied in the context of athletes and other highly skilled professionals (Schön, 1984), it has also been found to be an effective way for elderly people to ease their memory problems, and more (Berry et al., 1989; Doherty et al., 2011; Crete-Nishihata et al., 2012). However, there exists a limited understanding of how to apply lifelogging technologies to support senior citizens' reflections. Senior citizens are as diverse as other generations, but their familiarity with mobile devices is generally lower than found with younger generations. A survey reports that more people in younger segments in the senior citizen population use mobile devices than in the older segments (Ramón-Jerónimo et al., 2013). Therefore, it is unlikely that older people would all be comfortable carrying multiple sensors to collect various lifelog data. In addition, it is likely that their goals are quite different from those of athletes and other professionals (Schön, 1984; Ramón-Jerónimo et al., 2013; Vicente and Lopes, 2016). Even so, exploring effective ways to support senior citizens has been an important social issue in many societies with large ageing populations (WHO, 2015; United Nations, 2020).

This study uses applied thematic analysis methods, including an inductive approach, drawing on established, innovative, theme-based techniques to ascertain the practical feasibility and effectiveness of a lightweight GPS-based lifelogging approach among senior citizens in Japan (Guest et al., 2011; Harvey et al., 2016). Thus, a total of ten participants were asked to carry a smartphone when they went out, and these devices automatically captured their activity and location data for three months. To aid reflection on participants' daily lives, a personalised report was generated every month to summarise their activities. Analysis of the lifelogging exercise was based on the log data collected by the mobile devices, questionnaires administered to participants every month, the activity reports, and an interview with the subjects at the end of the study. As a result, this investigation in data collection is expected provides evidence to broaden the depth and scope of lifelogging research, to advance understanding of senior citizens and to inform policy interventions.

This article is organised as follows. The next part describes the literature in the field used to identify lifelogging by senior citizens; this is followed by our research questions. The subsequent section introduces the study design used throughout. A section describing the results is followed by a discussions section, and then, the study conclusions.

Literature

The literature on the study of lifelogging is sparse, and most work in this domain concerns the technical advances of relevant hardware or software (Shah et al., 2012; Gurrin et al. 2014). Early research on lifelogging focused on developing new types of sensing and display hardware (Mann, 2004; Gurrin et al., 2014). For example, Aizawa et al. (2004) presented a life diary recording system using video, audio, accelerometer, gyroscope, GPS, annotations, documents, web pages and email to create an index and retrieve the resulting image data. Aizawa et al. (2004) stated that to confirm the accuracy of scene retrieval from the records, other combinations of detailed contexts and contents must be used to reflect a person's interests. If this study requires more specific place identification, the location from a GPS would be used as the first step to specifying the exact setting of the activity. Next, these researchers would find conversation scenes at or near this exact place from the contents of the video data.

Shah et al. (2012) performed a content analysis of information gathered from various sources, and developed a system to archive and retrieve long audio recordings in a lifelogging scenario. They considered using multiple information sources to minimise the limitations of individual sources by combining multi-modal details, such as location, movement, audio and video, to characterise daily activities.

Lifelogging systems must be robust and unobtrusive because the human body is a harsh environment for sophisticated technology (Gurrin et al. 2014). On the other hand, Ruckenstein (2014, p. 68) leaned into the concept of the *data double* as "the conversion of human bodies and minds into data flows that can be figuratively reassembled for personal reflection and interaction." Their findings suggest that these data assemblies can create permanence and stability while profoundly changing the way people think about themselves, others, and their daily lives. Self-monitoring technology that can aid social research offers the possibility of crossing the divide between biology and society.

Lupton (2016) also had the idea study self-tracking activities from a sociological perspective, stating that there are several terms to describe "the practices by which people may seek to monitor their everyday lives, bodies and behaviours". These include self-tracking, lifelogging, personal informatics, personal analytics, and the quantified self. As for lifelogging, Lupton characterised it as "the specific practice of using wearable computing devices such as cameras, sensors, and other computerised and automated ways of collecting personal information over a period of time".

Lifelogging using lightweight GPS-based technology has been studied in several works (Aizawa et al., 2004; Hurvitz et al., 2014; Joho et al., 2016). For example, Hurvitz et al. (2014), identified that the integrative characteristics of large datasets contained in lifelogging software such as LifeLogs and SmartMaps hold great promise for advancing spatial epidemiologic research to promote and facilitate healthy behaviours. Joho et al. (2016) added that demographic characteristics influence the use of GPS in ageing societies.

Furthermore, lifelogging can benefit both older and younger people in various disciplines. For example, West et al. (2017) investigated the arena of digital health that involves the increased ubiquity of self-tracking practices by individuals. In turn, these are driven by the proliferation of self-tracking tools and technologies, which leads to using self-tracked data as evidence for clinical decisions. Jalal et al. (2014) proposed human activity recognition (HAR) for elderly monitoring applications, such as monitoring health problems or checking people's indoor activities at home, the office or in the hospital. In addition, Erikson stated in his stages of psychosocial development that people in late adulthood (60 years old and above) want to find a sense of balance and to reflect as their final developmental task (Erikson, 1950, 1968; Orenstein and Lewis, 2020). For example, this could include late adulthood contemplation and acknowledgement of personal life accomplishments (Orenstein and Lewis, 2020).

Another study by Obo et al. (2015) developed a visualisation system to represent personal relations between elderly people and their family members based on their daily activities. Visualisation as part of user lifelogging can help one understand and share personal preference and lifestyle (Yang and Gurrin, 2013; Obo et al., 2015). However, a limitation was identified from the previously mentioned existing works. For example, locus and time influence the users' behaviour that will be recorded by the lifelog system, yet they have not been examined through a qualitative assessment based on applied thematic analysis in Japanese senior citizens. Thus, lifelogging technologies can play an important role in addressing this problem, since it is now possible to gather long-term quantifiable self-data using advanced wearable devices and mobile applications.

Research questions

Lifelogging practices can have different impacts depending on different people's life stages (Orenstein and Lewis, 2020). For senior citizens, lifelogging technologies are expected to be effective in supporting their health and well-being. However, it is unclear to what extent existing lifelogging devices are designed for senior citizens, and the process could be overwhelming for them. Therefore, it is important to investigate and identify effective lifelogging practices for senior citizens.

This study had two main research questions:

1) Is it feasible for senior citizens to carry a smartphone to exercise lightweight lifelogging? What does a lifelogging practice by senior citizens look like?

2) How can we present the result of their lifelog data to facilitate senior citizens' reflection on their daily lives? Can senior citizens make sense of the analytical results of lifelog data? Where are the difficulties, if any?

By addressing these questions, this study aims to provide insights into the opportunities and limitations of current lifelogging technologies and services, to facilitate data-driven reflective learning by senior citizens.

Methods

Study design and ethical approval

The study design was approved by the ethics committee of the Faculty of Library, Information and Media Science, University of Tsukuba. It was conducted between April and June 2016; the season was Spring in Japan, with the average temperature ranging between 59.7 and 72.3°F (15.4 and 22.4°C). Due to the consent with participants, we report the results of behavioural data where individuals cannot be identified.

Participants

A recruitment process was as follows. First, we identified several local communities from the database provided by the city council. We shortlisted some communities to approach based on the two criteria: the age group (i.e., above 65 years old) and group size (i.e., over 20 members). One group who satisfied our criteria responded positively to our call for participation. A total of ten senior citizens, who were above 65 years old and physically and mentally healthy, participated in our study from the group. Of those, six were female, and four were male. The mean age was 75.0 (SD: 2.94). They all belong to the same local community organisation, which aims to perform gentle physical exercise every week. The recruitment of participants from this community was intentional for practical reasons, as it meant that we could easily keep in close touch with all of them and that the participants could help each other with minor technical questions (if any). It should be noted that their level of enthusiasm to participate in the study was generally high, although it changed over the three months. Given that this was the first study for us, it was our intention to obtain strong signals for better understanding of their practice. Investigation with other populations is left for future work.

Our face sheet that were completed by the participants identified the following additional profiles: The number of people living in participants' homes varied from zero to six. All participants either had a part-time job or regular volunteer work. As for existing recording habits, nine out of ten participants had a habit of writing a diary, six carried a step counter, and one measured blood pressure in the morning.

Device

We used a particular model of Android-based smartphones (FREETEL Priori3 LTE) in this study. The price was in the range of $100–120. Next, a GPS and activity-tracking app (Moves.app) was installed on each device to collect the data without manual operation by participants. In addition, the devices were set to disable notification functions, so they would not distract participants from their daily lives or interfere with their activities. A preliminary test conducted by the authors indicated that the smartphone's battery lasted from 18 to 24 hours in most cases. Therefore, we asked participants to charge the batteries every day before they went to bed.

Among the various potential devices available for lifelogging, our choice was intentional for two reasons. First, we wanted to start with a simple device since this was our first study asking senior citizens to practice digital lifelogging. Second, participants' physical and mental safety was our first priority, and thus, we avoided the potential overload of using more advanced technologies in this study. The use of more advanced technologies is left to our future work.

Orientation workshop

Before the data collection period, we invited potential participants to a workshop to introduce the study. In the workshop, we first explained the aim of the study with an information sheet and then asked participants to sign a consent form upon agreement. All agreed to participate. Next, participants were asked to fill out an entry questionnaire (face sheet) to provide their demographics, family structures, existing customs of recording their daily activities, and experience using smartphones.

Then, we handed a customised smartphone to each participant and explained the following using slides and a screen projector: 1) Using the home button; 2) Swiping on the screen; 3) Daily routine during the study (checking the status of the smartphone in the morning, taking one's smartphone when going out, charging the device before bed); 4) Automatic GPS data collected by the smartphone; 5) Warning about the use of smartphones on the street or near water; 6) Accuracy and limitations of GPS signals and their potential impact on the collected data; 7) Checking the GPS data using an app; 8) Data screening to identify dates to exclude from analysis; and, finally, 9) Contact information for any technical and other general issues.

A print-out of the slides was also given to participants during the workshop. The workshop ended with a Q&A session to clarify their questions and concerns about the study. The workshop lasted approximately two hours.

Data flow and visualisation

The flow involved in this study obtaining lifelog data from participants and creating monthly reports is illustrated in Figure 1. As shown, the original data from the participants' activities were obtained by the smartphones' GPS recording app. This app was set to periodically upload the data to secure cloud storage. This prevented losing valuable data even if the device were accidentally broken or damaged. We distributed a data screening form for participants to inform us of any dates they did not wish us to include in the analysis and reports. However, no participant requested the removal of dates or data during the study. This might have been due to the study design; that is, collected data were not shared with anyone other than the researchers. We then obtained the data from the cloud storage for analysis.

The tool chain used from the data analysis to the report generation is illustrated in Figure 2. First, we exported the data files from Moves app, and indexed them using a search engine, Elasticsearch. The visualisation was mainly generated by Kibana, which retrieved the relevant data from Elasticsearch for data analysis. Finally, a content management tool, Hugo (version 0.15), was deployed to generate individual reports by importing visualisations created by Kibana (version 4.4). Hugo's outputs were then printed and delivered to participants.

Figure 3 shows some examples of the visualisations we created from the lifelog data that we obtained from the data flow described above. Figure 3a shows the number of hours a participant moved outside (daily) for a particular month, and Figure 3b shows the breakdown of outside hours by day of the week as well as time of day. This example indicates that the participant went outside mostly between 6 a.m. and noon on Sunday, unlike on other days of the week. Figures 3c and 3d) show the geolocation of their movement on the map, where Figure 3c shows the entire region of their movement in the month to illustrate how far they travelled, and Figure 3d shows a detailed map of their movement in a particular region to help the participant remember each day. In the monthly report described next, these visualisations were cross-referenced where appropriate to facilitate participants' understanding of their behavioural patterns from different analytical data.

Monthly reports and questionnaires

One of the challenges we faced when utilising lifelogging technologies to facilitate senior citizens' reflection was the interpretation of quantified data. On this subject, a range of infographics has been proposed and implemented by others

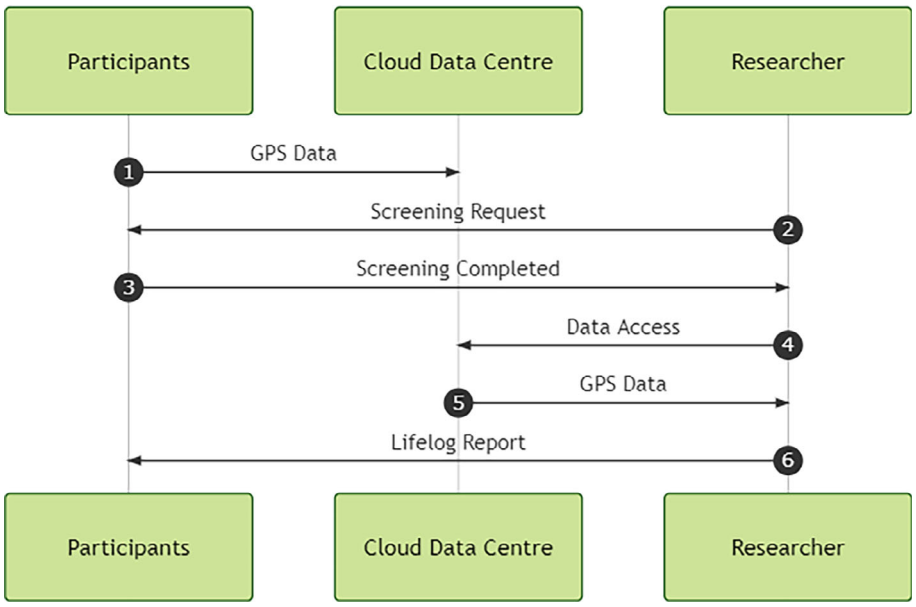

Figure 1. Data flow. GPS data were first obtained and stored by a mobile phone, and a copy was made in secure cloud storage. Researchers then analysed the data.

Figure 2. The tool chain used in this study to analyse and visualise individual lifelog data and generate a monthly report that included manually crafted comments.

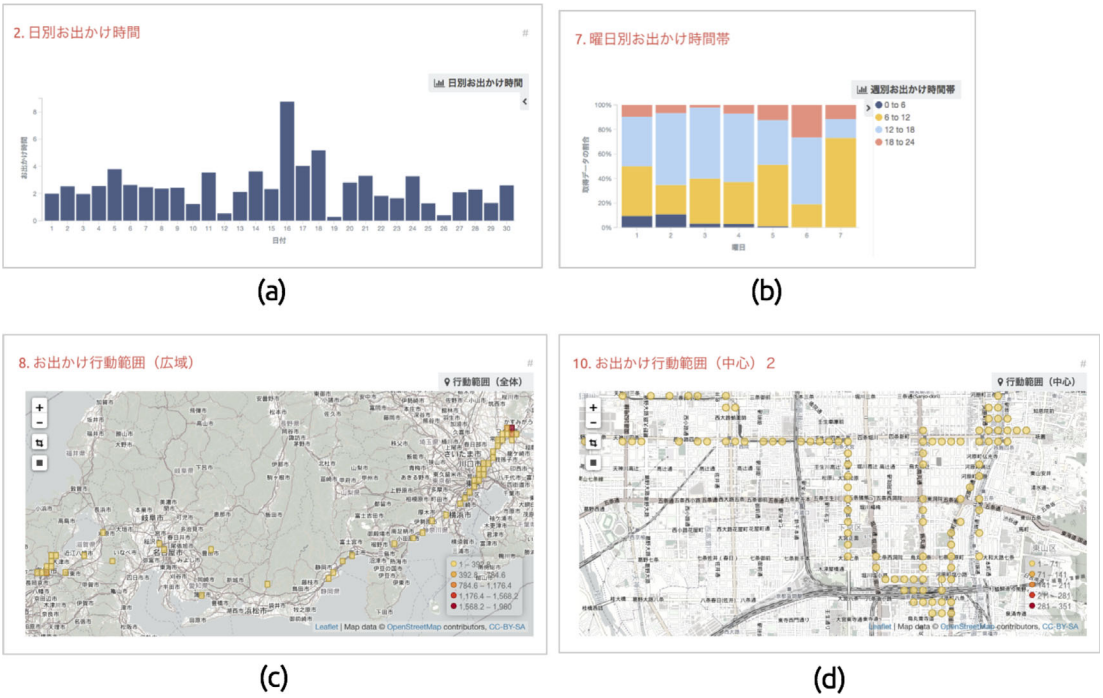

Figure 3. Visualisation of lifelog data. (See text for a description of each visualisation).

for the visualisation of lifelog data (Yang and Gurrin, 2013; Obo et al., 2015). However, the interpretation of visualisations in a study such as ours is still left to end-users, which can be difficult unless they are familiar with underlying data structures (Wilson et al., 2016). Therefore, we decided to generate a periodic lifelog analysis report, which included two pieces of information along with each visualisation: how to read the chart and an editorial review of the data trends.

Every month, we manually generated a report that summarised participants' activity data, and examples of our monthly report are found in Figure 4. Since this study focused on GPS data, the report contents were based on the locations the participants visited, transportation methods (e.g., walking, running, cycling, car, public transportation), and the time-stamps.

There were ten items of lifelog data analysis in the monthly report, and a summary of these results can be found in Table 1. There is little insight in the literature regarding the best practice for presenting the analytical results of lifelog data to senior citizens except for the sizes and colours of fonts and diagrams. Thus, we decided to adopt a simple structure to present each of the analytical results with three components: 1) number of diagram, 2) how to read the diagram, and 3) any pattern found in the diagram.

The numbers of diagrams were the presentations of the analytical results of participants' lifelog data, as visualised by the experimenter. The visualisation methods varied based on analytical viewpoints. Figures 3 and 4 show some of the examples of our visualisations. As can be seen, some data were presented as a bar chart, whereas others were presented as a pie chart. Geographic data were presented as a map. Some annotations such as those for travel methods were automatically determined by the application used in the study based on the speed of movements.

The second component in the monthly report was the description of how to read the diagram. The education level of the senior citizen population is diverse, and thus, we did not assume that participants knew how to read the numbers and diagrams. Therefore, we provided a brief description of what each diagram meant, what each colour represented, what a large proportion meant, and so forth.

The third component was the description of any pattern found from each month's data. This included highly personalised texts, individually tailored so that participants could interpret the outcome of the diagram. We also had a general policy to focus on positive aspects of the data rather than negative aspects, if there were any. However, the generation of the third

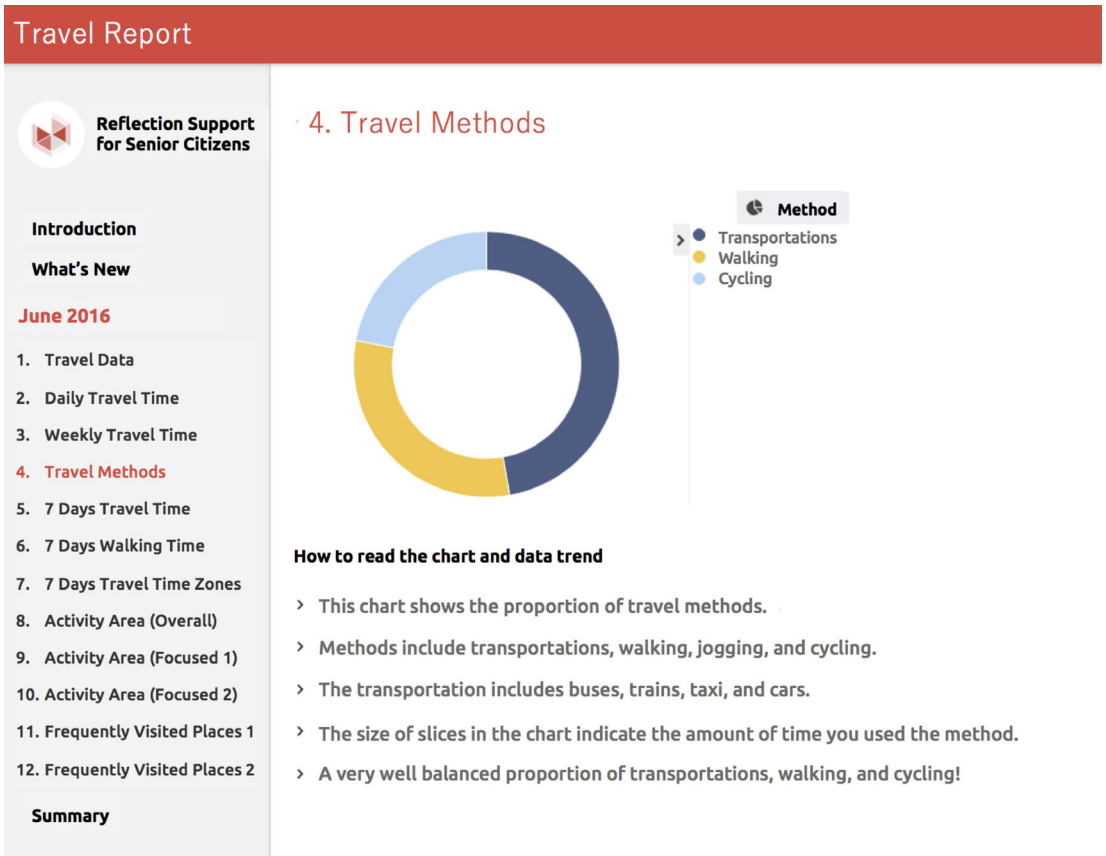

Figure 4. **Example page of the lifelog reports sent to participants once a month for reflection.**

Table 1. **Ten items visualised and described in the lifelog report.**

Report Item	Description
1. Travel Data	Travel data were collected, including the number of days, the number of places visited, and number of hours out
2. Daily Travel Time	Breakdown of travel hours per day; participants can observe dates, with the hours spent going out
3. Weekly Travel Time	Breakdown of travel hours per week
4. Travel Methods	Breakdown of travel hours per transportation method (e.g., walking, cycling, transportation). Transportation includes cars and taxis, and other public transportation such as buses and train
5. 7 Days Travel Time	Breakdown of travel hours per day of the week (Monday, Tuesday, Wednesday, etc.)
6. 7 Days Walking Time	Same as above but focus on walking
7. 7 Days Travel Time Zones	Breakdown of 7 Days Travel Time over four segments of time (i.e., 5am - 8am, 9am - 12pm, 1pm - 4pm, 5pm - 8pm)
8. Activity Area (Overall)	Visualisation of visited places on a map, to cover the entire region visited during the data collection period (month)
9. Activity Area (Focused)	Same as above but focuses on a local area where visited locations were concentrated
10. Frequently Visited Places	Points of interest on the map that participants visited frequently during the data collection period (month)

component was highly contextual, and thus the development of systematic interpretation texts will be included in our future work.

In addition, the monthly reports for months two and three contained comparison to previous months. The comparative results aimed to highlight any changes observed between the current month and the previous one. Thinking about the cause or reason for the difference could be an effective trigger for participants to reflect and characterise a month's activity against previous months.

Beyond this, a questionnaire was attached with the monthly reports sent to participants. The questionnaire consisted of four sections, capturing participants' perceptions of 1) the overall design of reports, such as font size, ease of colour recognition, and the number of figures; 2) the ease of understanding results and the explanations of each result, including any reflective thought that emerged from each of the results; 3) any changes in lifestyle or new awareness in the past month; and 4) any other comments (optional). Note that, the original reports and questionnaires were generated and answered in Japanese, and English translations given in this article are for reference only.

Technical support

Since most of the participants did not have previous experience using smartphones, they faced technical issues now and then. They were able to contact us via phone when they had such problems. We received several reports about technical issues in the first two to three weeks. Common problems included: 1) Software updates/notifications (the mobile operating systems made several notifications even if configured to be quiet, and this confused participants); 2) Battery issues (when GPS logging is activated, it uses far more battery power than usual, so sometimes the smartphones ran out of power); and 3) Flight mode (some participants accidentally activated this, which stopped all sorts of connections, but it was not very obvious to them).

However, the number of reports began to decrease rapidly by the time the first data screening was conducted, which was approximately four to five weeks from the beginning of data collection. We also observed that the frequency of technical issues was relatively skewed toward certain devices, which could indicate that the occurrence of technical issues was due to a mixture of factors, such as the device manufacturing quality, participants' lifestyles, and participants' experience of operating smartphones. Furthermore, some participants were more interested in exploring features and functions available on smartphones, whereas others did not pay much attention to the additional functions.

Final meeting and questionnaire

After the report for the final month was sent to the participants, we invited all of them to attend our final meeting session and asked them to express any thoughts or feelings about their participation in the study and their lifelogging practice. This was a group session which allowed us to observe discussions among participants. The final meeting lasted approximately 60 minutes. In addition, we asked participants to fill out a final questionnaire for summative feedback on the various aspects of the study. The areas covered included: 1) The burden of carrying a smartphone when they went out; 2) The burden of charging the device every day; 3) Ease of learning how to operate the device user interface (UI); 4) The precision of recorded data; 5) Ability to check their behavioural data at any time of the day; 6) Frequency of checking their data in a day; 7) Impact of carrying the device on their daily life behaviour and patterns; and 8) Reactions on receiving personalised monthly reports.

During the final meeting, three of the study authors were taking notes to capture feedback from participants. Our recording of the meeting was also cross-referenced with our notes to ensure that we did not miss or misinterpret the comments from participants. We present the findings from the final meeting in the discussion section.

The original questionnaires were submitted by participants using handwritten texts, which were re-typed into a spreadsheet (Microsoft Excel for Mac 2011), and one of the authors went through for the first coding to manually identify clusters of subjective assessments using an affinity diagram method. Another member of the team independently validated the integrity of the coding and the structure of the diagrams. Then, two of other authors validated the structure of clusters by resolving the cases where clarification was needed.

Results

This section presents the findings from our analysis of lifelog data, questionnaires, and the meeting session.

Lifelog data

The GPS tracking app installed on the smartphone of ten participants captured a total of 1,547 data points during 1,215 hours of outgoings over the course of three months. We noticed that missing data points were fairly common during

the study, which could be caused by several factors, such as bad reception of GPS signals, loss of internet connection, participants forgetting to take their smartphones when they went out, or their batteries running out during the day. Therefore, the number of data points we collected should be seen as a lower boundary of the data size. Nevertheless, the large number of data points collected provides a great opportunity to understand the participants' daily activities and patterns. Figure 5 shows three levels of granularity that demonstrate the accuracy of our GPS data. As you can see that, the GPS tracker in our mobile phones was reasonably accurate in capturing a position at the street level.

Figure 6 shows the total number of hours recorded for ten participants during the three months of data collection. The number of hours varied significantly over time, but the trend line using a 7-days moving average suggests that the number slightly increased every month, suggesting that the lifelogging and lifelog-based reflection can encourage senior citizens to go out. The GPS data also showed that our participants went out with a similar level of frequency across the days (Monday to Sunday) but the frequency of going out in the morning was found to be higher from Friday to Sunday. There were three groups of participants regarding their main means of moving: 1) Mostly by car; 2) Half car, half walking; and half car, 1/4 walking, and 1/4 cycling. Although there were other aspects of individual GPS data that can be analysed in-depth, we leave that to our future work.

Reflective thoughts from lifelog data

The monthly questionnaires asked participants what they found from each visualisation and its description, as well as what reflective thoughts they provoked. During the coding and affinity diagram generation, it became clear that it would

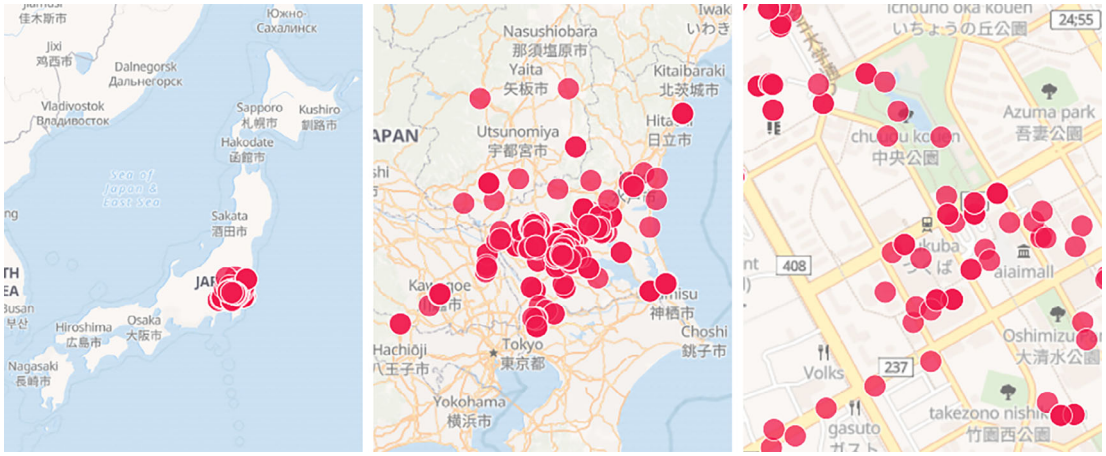

Figure 5. Recorded GPS points at three levels of expansion. From left: Japan, Kanto Area, and City Center. One can see that the GPS tracker of our mobile phones is reasonably accurate in capturing a position at the street level.

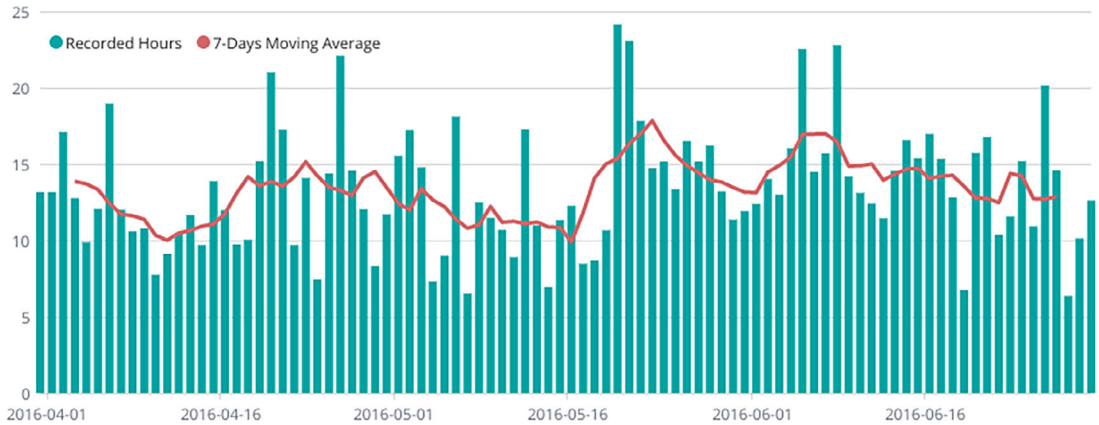

Figure 6. Total number of hours recorded for ten participants during the three months of data collection. The number of hours recorded varied significantly over time, but the trend line using 7-days moving average suggests that the number slightly increased every month.

be best to employ two dimensions to better organise participants' reflective thoughts: Lifelogging Process and Affective States. The former had four categories: 1) Lifelogging practice, 2) Monthly reports, 3) Interpretation of lifelog data, and 4) Thoughts and actions about the findings. On the other hand, the latter has three categories: 1) Positive, 2) Neutral, and 3) Negative. This section presents the findings using these two dimensions.

Lifelogging practice

The first category of responses we identified was on the lifelogging practice using a smartphone. At an early stage of the study, we observed responses such as a short battery life of the smartphone and an additional workload from charging the device every night. The participants' affective states on these responses were negative, which seemed to be due to the adaptation to a new practice and device. At a later stage of the study, we identified responses regarding their adaption to the new practice (e.g. "It has become a habit to bring a smartphone with me when I go out"), which could be seen as a positive state. In addition, some responses described the lifelogging practice's impact on behavioural/affective change (e.g. "It has become enjoyable to go out").

Monthly reports

The second category we identified was a set of responses on monthly reports. This category included the responses to the layout and looks of reports ("It was difficult to see the difference of colours used in the diagram"), suggestions for the report design ("Why don't you show this diagram in this way?"), and the recipient of the reports ("I look forward to receiving the next report"). As in the previous category, one can see that these responses could also be viewed from either positive or negative perspectives. This and the previous category had a distinct difference from the rest of the responses to form a group, but the size of the responses was relatively limited.

Interpretation of lifelog data

The third category includes a range of responses regarding participants' interpretations of lifelog data presented in the monthly reports. From the affective state perspective, many responses in this category were neutral. Examples of neutral responses include the description of behavioural patterns ("This was the month where I used a car a lot"), explaining the aim of a behaviour ("I went there because of my work"), explaining the reasons for their behavioural patterns ("I refrained from doing this due to my condition"). Meanwhile, negative responses often concerned a difficulty in understanding or finding a pattern in lifelog data ("I don't understand this diagram"), or the poor precision of captured data ("I was disappointed by the fact that there were missing data in my report", "Datapoint was too small to tell anything"). On the other hand, positive responses occurred when participants found patterns in the data ("I can see my behavioural patterns well from the report", "The locations where I visited were recorded with an incredible level of accuracy"). In the second and third months, the diagrams of the previous month were presented to encourage participants to interpret the current month's data based on the difference from or similarity to the last month's data. Some participants even compared the data in the monthly reports with their own recording of daily activities.

Thoughts and actions upon findings

The last category included participants' thoughts on the findings from the lifelog data. The most basic responses in this category were reflections on their behaviour ("I didn't do much walking last month", "I wish I could do without a car, but I can't"). At times, the lifelog data prompted action from participants ("I revisited my diary since the number of this diagram was very high"). They also expressed differences in their expectations ("I thought I would find this colour more frequently on the map"). Some participants compared their patterns to previous reports ("The number increased because I avoided using cars", "The accuracy of data capture was better in the previous report"). Finally, this category included responses about their resolutions and wishes for future behavioural changes ("I would like to walk more").

Ease of use of monthly lifelog reports

Every month, we asked participants to indicate how easy it was to read the contents of the reports, including the description and result of each visualisation. A five-point Likert scale was used to capture participants' perceptions, where 5 indicates a strong agreement to a statement, such as "It was easy to understand the description of Item N", and 1 indicates a strong disagreement with the statement. The boxplots of the results are shown in Figure 7(a) and 7(b) for the description and results in the report, respectively.

Figure 7(a) suggests that participants found it easy to read the descriptions of numbers, figures, and maps. As can be seen, the result suggests that our manually generated personalised messages facilitated participants' understanding of their

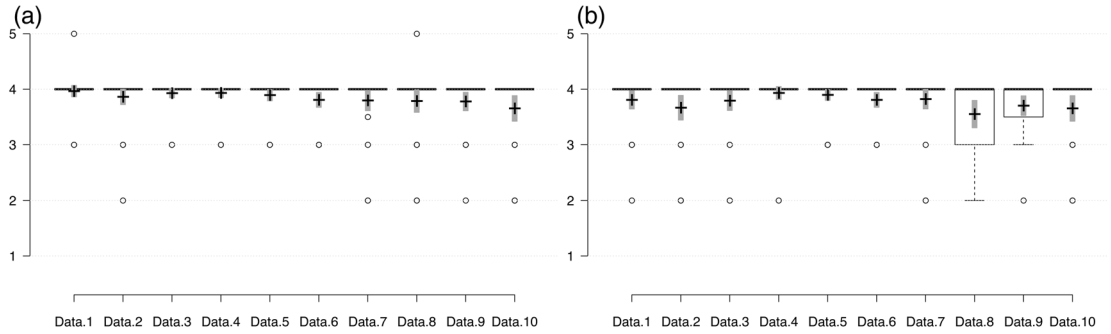

Figure 7. Ease of use regarding the description of ten visualisations (a) and their results (b) over the three months (N=30). Data ID corresponds to the report item in Table 1.

lifelog patterns and their implications. Figure 7(b) suggests that participants found it easy to read the results from the visualisations. Again, the results support the effectiveness of our monthly reports. However, some participants found the map visualisation of lifelog data less easy to interpret. One reason for this is that the lack of data was more visible in the map representation than other diagram-based representations. Therefore, participants noticed missing GPS points of places they remembered visiting. They were less likely to notice discrepancies for the accumulating statistics of time and frequency given in other items.

Other feedback from the monthly lifelog reports

A questionnaire was attached to the monthly lifelog reports sent to the participants to capture their perceptions of the analytical results presented there. A total of three reports were sent to the participants from April to June 2016. Some of the highlights in the participants' feedback included the following:

First, participants were generally more positive about geographical visualisations of lifelog data, such as activity areas and frequently visited places, than about activity time-based visualisations. One reason seems to be that many participants already had a habit of carrying a simple step counter with them. Therefore, the information generated by activity time had a lower level of novelty to the participants. Second, the visualisation of travel methods seemed to have an impact on their understanding of lack of walking in day-to-day life. Many of them somehow sensed that they relied on cars and public transportation too much, but the visualisation of the proportion of travel methods reinforced their 'guesses'. As a result, many participants commented that they would try harder to walk more. Third, the instructions on how to read charts and interpretations in the reports were well received by participants, and most stated that the visualisations were easy to read and understand. When one gave a low score for understanding a chart, it was often due to a mismatch between their memory and the visualised data. Such a mismatch could stem from multiple factors, including inaccurate data positioning by budget smartphones. Overall, the lifelog report was assessed to provide new insights into the participants' lives that had not been offered by conventional data collection methods such as diaries or simple step counters. We plan to analyse how their perceptions of daily activities change over the multiple months of the study.

Feedback from final meeting

A final meeting was held four weeks after the final monthly report was sent to participants. The final meeting aimed to capture participants' overall perceptions of the lifelogging exercise and any suggestions to improve the practice.

Positive findings

The participants' ability to see their activity area on the map was a source of frequent positive commnets This was a new experience for many, and the feedback during the final meeting indicated that this was a great way to reflect on their days or highlights of the month. In addition, other participants indicated that some of the activity-based lifelog data was similar to that obtained by the step counter, which most participants had experience of carrying and checking. On the other hand, the map visualisation of their activity trails provided related but different data to participants. This demonstrates the advantage of using GPS-based sensors in addition to conventional movement-only sensors.

Another piece of positive feedback on the monthly reports concerned the ability to see distance and activity time. Again, step counters could show only the total amount of walking they performed, but our device was able to capture movement by other transportation methods, such as bicycles, cars, and public transportation. Furthermore, the monthly reports included a breakdown of the activity over five segments of each day, which allowed participants to visually understand when they were actively going out during the day. Although this was slightly complex data to interpret, participants found

it useful, partly because they could see the pattern of their daily lives over the days of the week. One participant mentioned that it was comforting to find that the monthly reports qualitatively confirmed their understanding of daily life.

Negative aspects

In addition, participants expressed negative aspects of their experience. The most frequent comment was about the poor precision of recorded data. The precision of GPS data can be degraded by poor reception of the signal from GPS satellites. Tall buildings in the surroundings, the first few minutes after leaving a house or building, and the way one carries a smartphone (e.g. in a pocket, deep in a bag) all affect the reliability of GPS signals. When the device's perception of GPS signals was poor, the recorded data could be misleading. For example, the device might record a location or route that participants did not use, or the device might miscalculate transportation methods (cycling rather than driving) because the GPS-based apps tend to estimate the transportation method based on the speed of movement. This estimation could be wrong when the base location data were poorly recorded.

Participants showed a great level of attention to the difference between the data of the device and their step counters and their memories. When the precision of location data was accurate, we received a different comment, which was a feeling of creepiness coming from the visualisation of precise movement that participants made throughout the day. One participant even said that it felt a little scary.

On lifelogging activities

Since this was most participants' first time using a smartphone, we asked whether or not carrying a smartphone and daily charging were a burden for them. The responses show that most participants did not find carrying a smartphone too burdensome, but some participants found charging every night a burden. This could be because the battery life of conventional mobile phones tends to be much longer than that of smartphones in general. Furthermore, tracking a GPS signal at most times of the day consumes a lot of battery, and thus, smartphones had to be frequently charged in the study.

A touch-based operation in the typical user interface (UI) of a smartphone is different from a physical, button-based operation in conventional mobile phones. This can be challenging for senior citizens. Although we spent a good amount of time playing with the touch-based UI during the workshop, some participants found it difficult to learn how to operate them. A smartphone is a complex device, and sometimes during the study, one gave unexpected notifications or forced system updates, which was beyond the scope of skills participants were asked to use in the study. As discussed earlier, many participants found that the precision of recorded data was occasionally poor. However, given that some people rarely had precision issues, this could have been due to the geographical environment or conditions of their activity areas.

Despite occasional problems with the recorded data, participants were mostly happy with the ability to check and monitor their activity data at any time of day.

The next set of questions investigated to what extend the participants' lifelogging changed the way they spend their daily lives. Ultimately, most participants were able to spend their daily lives as usual, and the start or end of the lifelogging exercise did not to a large extent affect the number of times they left their homes. Two participants said they started a new habit since they participated in the study. Both started walking when they were not at work, or in the morning with their partner.

Finally, we asked about the participants' overall satisfaction with monthly reports and lifelogging services in general. Half of the participants indicated that they looked forward to receiving monthly reports, and they would participate in a similar service in the future. These results can be taken either positively or negatively. In a positive sense, collecting GPS-based lifelog data successfully captured aspects of their daily lives, and monthly reports supported reflection on their daily lives. In a negative sense, the exercise did not provide enough benefits to participants to make them feel this could change their lives.

Summary

The results of the analysis suggest that 1) It is feasible for senior citizens to carry a smartphone to collect their activity and location data, and participants did not feel stressed even when they did not have previous experience using the device; 2) The activity report is an promising way to help senior citizens reflect on their daily lives; 3) Senior citizens can be very careful about erroneous numeric data captured by these devices; and 4) The ability to interpret visualisations of lifelog data can vary across participants.

Discussion

This section discusses the implications of our findings for the practice of senior citizens' lifelogging.

Feasibility of a lightweight GPS-based approach

Our first research question concerned how to design a lifelogging practice tailored to senior citizens and investigated the feasibility of adopting lightweight GPS-based approach using smartphones. As discussed earlier, some participants had technical issues at the early stage of the process, and thus, the time taken to incorporate the smartphones into their lives varied. Having said that, all of the participants said that carrying the devices, checking the data once a day, and charging them regularly were not significant burdens in their daily routines. However, due to the increased use of battery life when GPS logging is activated, we recommend providing dedicated devices for research purposes. Although this meant that sometimes the participants in this study had to carry two devices (one of their own and one from the researchers), this prevented us from interrupting the use of their own devices, which could cause significant problems if they were far away and needed help. In addition, by providing dedicated devices in such a study, one can ensure that non-related apps are not interfering with the lifelog data collection.

In previous research, Harvey et al. (2016) and Gelonch et al. (2019) revealed that battery life does not affect older adults using lifelogging wearable cameras. This is because a wearable camera has a different function than a cell phone. However, not surprisingly, having two smartphones makes senior citizens more aware of their body's reactions and daily actions. Self-tracking will be more pervasive among those with the skills and means to connect with their bodies, minds and lives in data-driven ways. Smart devices including smartphones can promote a new framework for approaching normality and pathology in everyday life.

Effectiveness of data-driven presentation of lifelogging outcomes

Our second research question was about the impact of the data-driven presentation of lifelogs on participants' perceptions of daily activities and behaviours. We had several signals to suggest that the lifelogging activities had a positive impact, motivating them to improve their daily behaviour. For example, maintaining a good level of walking in their everyday life is literally a critical aspect to their well-being. Many participants expressed their reflection on the low level of walking activities found in the monthly report and their determination to increase the walking time. It takes time to change our behaviour and habits, but the overall trend of increasing time going out over the three months might be an indication of such changes encouraged by the lifelogging activities.

Another observation gleaned from this investigation was participants' strong attention to the numbers presented in the mobile phone app as well as the data shown in our monthly reports. As we show in Results section, participants expressed negative feelings strongly when the data presented in the app or reports were not accurate, and expressed positive feelings when the app managed to capture their patterns precisely. This is not unsurprising, or obvious. This suggests that by leveraging lifelog data obtained from their own everyday life, one can provide great opportunities to develop senior citizens' data literacy.

Limitations

There were several limitations in this study. First, the range of lifelog data one can collect is diverse, and this study focused on only temporal-location data. Although this allowed us to implement a lightweight lifelogging approach, which was suitable for senior citizens, the effectiveness of other kinds of lifelog data should be investigated. Our approach means that we collected participants' outdoor activities. A follow-up work to integrate with indoor activities is under way.

Second, given that lifelog data was obtained from individual participants' everyday lives, the findings and implications were limited to the environment where participants of this study lived, which was a particular area in Japan, although their lifestyles were quite diverse. Given that most participants had a daily habit of keeping a step count or diary, this behaviour could affect one's findings compared to people who did not have such a custom. Also, we did not investigate the sharing of lifelog data among the participants' friends as Brewer and Piper (2016) suggested.

Finally, although the lifelog technologies allowed us to collect detailed behavioural data, which would be difficult by conventional data collection methods in the social sciences, this research was not meant to be an ethnographic study. Some qualitative data were obtained from participants to understand the context of their behavioural patterns, but more qualitative studies would be needed to fully understand them.

Conclusions

This article has presented our investigation of an effective lifelogging environment that could be used by senior citizens. We proposed and implemented a lightweight lifelogging approach based on GPS data collected by a smartphone. A total of ten senior citizens participated in our study, and each collected her or his data for three months. A personalised report was generated and presented to the participants every month, encouraging them to reflect on their activities and their patterns.

This study demonstrated that senior citizens could use smartphones for lifelogging purposes after a time of adaptation. Although missing data are common, these devices can collect data and capture a good portion of participants' daily activities. The data-driven lifelog report was also found to be useful for participants to remember and reflect on their past activities. In addition, we made several observations of how such lifelogging exercises encourage senior citizens to expand their activity ranges. We note that the behavioural changes observed in this study could be due to many factors, such as weather, social relations, or health conditions. Further studies with a control group will be needed to gain a more comprehensive understanding of senior citizens' lifelogging practice and its impact on multiple aspects of their everyday lives.

Future work

There are two major directions suggested by the line of research of our study. One is to develop a more advanced framework for generating lifelog reports for senior citizens. In this study, we manually generated personalised comments for each of the lifelog data diagrams presented in the report. This is a highly time-consuming, skilled task. Ideal commentaries on the diagrams should have a tone of encouragement, rather than just describing the data pattern accurately. On the other hand, some basic description of analytical results might be automatically generated. Therefore, to create an effective lifelog report from data-driven results, we need to develop a framework to guide us to produce consistent yet encouraging descriptions of findings from the data. This could be a combination of manual and automatic operations using NLP tools such as Named Entity Recognition or more advanced language generation models.

Another direction would be to develop a framework for a data literacy learning program for senior citizens. Senior citizens' educational backgrounds tend to be more diverse than those of younger generations. Some participants had years of experience working with numbers and diagrams in their careers and thus were better at interpreting patterns and implications from the data presented in the reports. Others could find it challenging to extract semantic meaning from numbers and diagrams. However, given the development of data-driven and AI-based services in many areas of society, it is not unrealistic to assume that the opportunities to face data-driven information could increase in the future. In such a situation, developing senior citizens' data literacy could be an important part of the research agenda. This study demonstrated that participants show a strong interest in their own lifelog data, and thus, this could work as ideal learning material for data literacy programs.

Data availability

Underlying data

Open Source Foundation: Lifelogging Practice of Senior Citizens. https://doi.org/10.17605/OSF.IO/2YF5K (Joho, 2022).

This project contains the following underlying data:

- question_description.tsv: Question items for the evaluation of monthly report description, used for Figure 7(a)
- answer_description.tsv: Answers for the evaluation of monthly report description, used for Figure 7(a)
- question_result.tsv: Question items for the evaluation of monthly report results, used for Figure 7(b)
- answer_description.tsv: Answers for the evaluation of monthly report results, used for Figure 7(b)

Data are available under the terms of the Attribution-NonCommercial-NoDerivatives 4.0 International licence (CC BY-NC-ND 4.0).

Free answers to the questionnaires (in Japanese) and lifelog datasets that underly the results cannot be sufficiently de-identified and thus, they are not available to share for ethical and privacy considerations. A reviewer may apply for access to part of the datasets by contacting the corresponding author (hideo@slis.tsukuba.ac.jp) directly. Access will be available upon consideration of the request.

Acknowledgments

We thank our participants of the study for sharing their experience.

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✓ **Birger Larsen**
Department of Culture and Communication, Aalborg University, Aalborg, Denmark

This interesting article reports on a study that introduces a small group of senior citizens to lifelogging, based on lightweight smartphone GPS devises, with the purpose of investigating if lifelogging can help them reflect on their lives in general and exercise and movement in particular.

Strengths:
- Careful study design, including
  - well-chosen low-cost devices
  - careful instructions and follow ups
  - technical assistance in place
  - well-timed interventions and data collection points
  - and well-crafted lifelogging summaries to elicit reactions.
- Interesting results. albeit on a small sample

Weaknesses:
- Very small sample of 10 senior citizens
- Too small sample to provide any clear statistics on differences between the participants.
- Figure 3 labels are in Japanese – could be translated and overlaid.
- The data was collected in 2016, and much has happened technologically since then.

Overall, the study is small but solid. It is clear what was done, what the limitations are and the authors do not over-generalise. Given this, the presented results are interesting.

Is the work clearly and accurately presented and does it cite the current literature?
Partly

Is the study design appropriate and is the work technically sound?
Yes

Are sufficient details of methods and analysis provided to allow replication by others?
Yes

If applicable, is the statistical analysis and its interpretation appropriate?
Not applicable

Are all the source data underlying the results available to ensure full reproducibility?
Yes

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Information Behaviour, Information Retrieval, Interaction Design, Bibliometrics, Research Evaluation

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Reviewer Report 24 January 2024

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Margaret Currie
James Hutton Institute, Aberdeen, Scotland, UK

This was an interesting study which examined the feasibility and effectiveness of older people using a light-weight GPS lifelogging approach. It was an interesting study and generally well written but I do have a few concerns:
1. The majority of the literature cited is from earlier than 2016, I suspect that there will be more recent literature that has not been cited, thus it is difficult to establish if there is still a gap in the literature unless it has been updated (which is why I answered partly).
2.The study itself is 7 years old. Attitudes to and perceptions of technology have changed a lot over the last 7 years, particularly because of the COVID-19 pandemic. This is not mentioned as a limitation. Which is why I stated partly for the study design.
3. Although a lot of data were collected this was for a small number of participants, it is difficult to make quantitative generalisations on such a small number of participants.
4. The authors also state that participants seem quite active - they take part in a group as well as either working or volunteering. How representative are these individuals of the wider older population in Japan.
5. Figure 2, I don't understand this and it is not adequately described.

6. Figure 3: legends are in Japanese and are too small to interpret.
7. Title of figure 7 isn't complete
8. Positive findings section - correct spelling mistakes and grammar.
9. But generally I think the paper is quite descriptive, it needs to be more analytical, making it clear how it moves the literature and academic thinking on, I would also expect the discussion to refer back to ideas on lifelogging and how the paper is advancing or adding to them.
10. I answered partly with regards to the reproducibility because the authors have had to anonymise participants details; which I fully understand and fits with the ethical consent they got to undertake the study.

Is the work clearly and accurately presented and does it cite the current literature?
Partly

Is the study design appropriate and is the work technically sound?
Partly

Are sufficient details of methods and analysis provided to allow replication by others?
Yes

If applicable, is the statistical analysis and its interpretation appropriate?
Not applicable

Are all the source data underlying the results available to ensure full reproducibility?
Partly

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Older people, health, geography, digitalisation including attitudes and acceptance towards health technologies.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

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